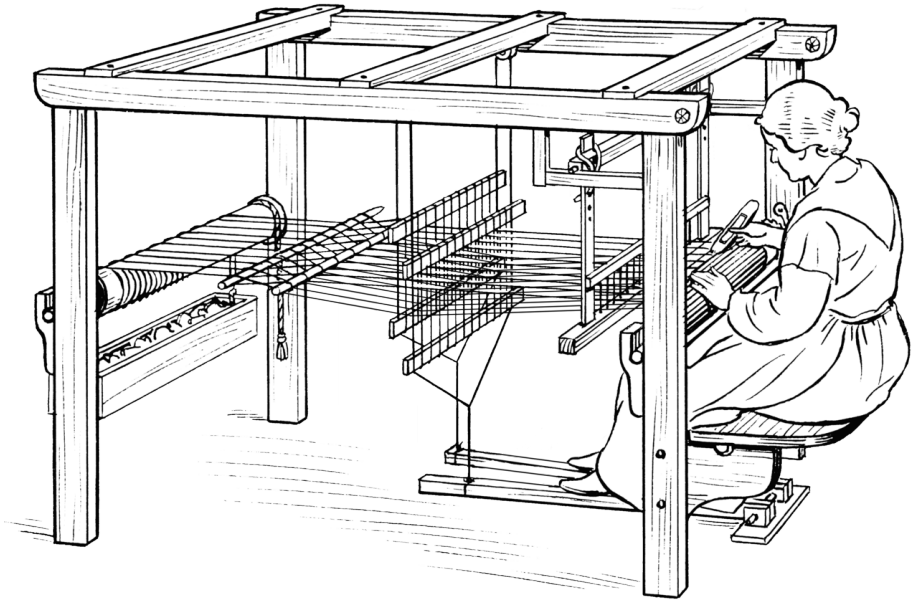
MathAlgorithm
A General Purpose Math and Algorithm Library



Introduction

The MathAlgorithm library provides commonly used math and algorithms used in Computer Science education in the C++ language. This book provides the documentation of the Library. Images, diagrams along with mathematical concepts are provided throughout.

Sort Algorithms

Insertion Sort

The library provides and **Insertion Sort** implementation through the `insertion_sort()` function.

```
void insertion_sort(std::vector<Data>& arr)
{
    for (std::size_t i=1; i < arr.size(); ++i) {
        insert(arr, i, arr[i]);
    }
}
```

The `insert()` function is implemented as,

```
void insert(std::vector<Data>& arr, std::size_t pos, Data& value)
{
    std::size_t i = pos - 1;

    while ( (i >= 0) && (arr[i] > value) ) {
        arr[i + 1] = arr[i];
        --i;
    }

    arr[i + 1] = value;
}
```

The **Insertion Sort** algorithm can be described as,

The complexity of the insertion sort algorithm is $O(n)$ for best case, $O(n^2)$ for average case and $O(n^2)$ for the worst case. The complexity is driven by the single while loop in the worst case,


```
        --i;
    }

    arr[i + 1] = value;
}
```

Graph Algorithms

Graph algorithms require data structures to represent input, intermediate, and output data. The library provides various data structures for the user to use to represent graph data. Here are the various data structures available in the library.

`Edge` Data Structure

The `Edge` data structure is used to represent an edge `(u,v)` between two vertices `u` and `v` in a graph. It is the base type that has different derived types depending on usage.

```
Edge
+u: int
+v: int
+weight: float
```

The starting or first vertex is identified by a unique integer value of u, and the ending or next vertex is identified as v. A weight is also provided to allow for edge weighted graph representation. A vertex can have many edges in a graph, so the value of u can appear many times. A sequence of edges could be represented by $(u_0, v_0, w_0), (u_0, v_1, w_1), (u_0, v_2, w_2)$.

`Graph` class

A graph is represented as a `Graph` class which encapsulates the graph data and provides unary operations, getters and setters.

```
Graph
-maxVertices: int
-directed: bool
-edges: list<Edge>
+Graph(int maxVertices, bool directed = false)
+load(std::string_view fileName): bool
+isDirected(): bool
+numVertices(): int
+isEdge(int u, int v): bool
+edgeWeight(int u, int v): float
+addEdge(int u, int v, float weight): bool
+removeEdge(int u, int v): bool
+friend operator<<(std::ostream& os, const Graph& graph): std::ostream&
```

An simple example of using the graph and an external json file to load is,

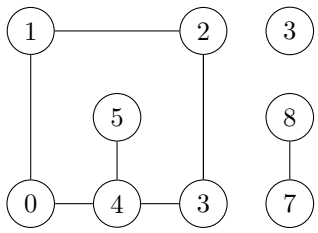
```
        Types::Graph graph;

        graph.load("data/test.json");
        std::cout << "\tOutput: ";
        std::cout << graph;
```

The output is,

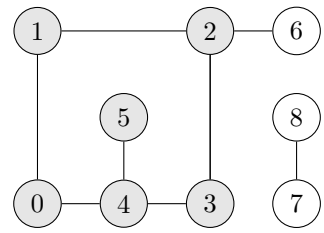
```
Output: { (1,2),(2,3),(1,4), }
```

And the json file is,

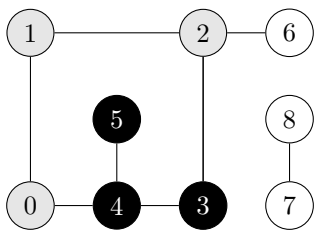
```
[
    {
        "u" : 1,
        "v" : 2,
        "weight" : 1.0
    },
    {
        "u" : 2,
        "v" : 3,
        "weight" : 0.5
    },
    {
        "u" : 1,
        "v" : 4,
        "weight" : 1.0
    }
]
```



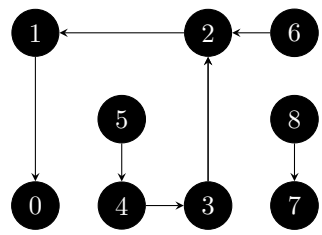
(a) Initial graphs state



(b) _dfs_visit() of each node coloring



(c) _dfs_visit() with some finished vertices



(d) _dfs_visit() final state

Figure 2: Depth-first search example

The `Algo::dfs(const Types::Graph& graph, int startId, std::vector<int>& predecessor)` function provides the `DEPTH-FIRST SEARCH` algorithm on a graph object and outputs a predecessor list to determine the paths taken back to the start for each child vertices.

```
void dfs(const Types::Graph& graph, int s, std::vector<int>& predecessor) {
        std::vector<int> vertices = graph.getVertices();
        std::vector<Types::VertexColor> color(vertices.size());
        std::vector<int> discovered(vertices.size());
        std::vector<int> finished(vertices.size());
        std::size_t counter = 0;

        std::fill(color.begin(), color.end(), Types::VertexColor::White);
        std::fill(discovered.begin(), discovered.end(), -1);
        std::fill(finished.begin(), finished.end(), -1);

        _dfs_visit(graph, discovered, finished, predecessor, color, s, counter);

        for (const auto v : vertices) {
            if (color[v] == Types::VertexColor::White) {
                _dfs_visit(graph, discovered, finished, predecessor, color, v, counter);
            }
        }
    }
```

The `Algo::_dfs_visit(const Types::Graph& graph, std::vector<int>& discovered, std::vector<int>& finished, std::vector<int>& predecessor, std::vector<Types::VertexColor>& color, int id, std::size_t counter)` provides the recursive calls to walk the entire graph. The `discovered`, `finished`, `color` and the final ouput `predecessor` lists are passed through each `_dfx_visit(...)` call. These are free functions and not a class object. This implementation is for an explicit example of where the affected data structures are used.

The pseudo code for the `DEPTH-FIRST SEARCH` is shown in listing,

Algorithm 1: How to write algorithms

```
Data: this text
Result: how to write algorithm with LaTeX2e
initialization;
while not at end of this document do
    read current;
    if understand then
        go to next section;
        current section becomes this one;
    else
        go back to the beginning of current section;
    end
end
```

An easier method of supplying a depth first search algorithm on a graph object is to wrap the operation in a class object to hide the implementation details. That can be seen in the `Algo::DepthFirstSearch` class that takes in a graph object and returns the predecessor list in a search function.